COMSATS University Islamabad (Lahore Campus)

Assignment <1>

Course Title:	Game Development	Course	CSC495	Credit Hours:	3
Course Instructor/s:	Aqsa Nawaz	Program Name:	BCE		
Due Date:	**11:59 pm 22nd Feb 2026**	**Maximum Marks:**		**30**	

Important Instructions:

- Do your Assignment in new Unity Project
- **Paste 5 screenshots of your game and cs scripts as in Assignment_Submission File.doc(attached) and submit as your registration_number.doc , like FA22-BCE-00.doc**
- **Also upload a 10 seconds video**
- DO NOT COPY YOUR WORK FROM ANY SOURCE
- Submit your Assignment at Google Classroom Assignment 1 folder

Question **Marks: 30**

CLO: : <2>; Create different assets and scenes for a game scenario.

Assignment 1 — Unity 3D Scene

Objective

The purpose of this assignment is to evaluate your understanding of Unity fundamentals including creating a project,

building a scene using built in 3D objects, applying materials/textures, using physics properties, and attaching scripts.

Assignment Task

Create a new Unity 3D project and design a simple meaningful scene demonstrating your basic grip on Unity tools.

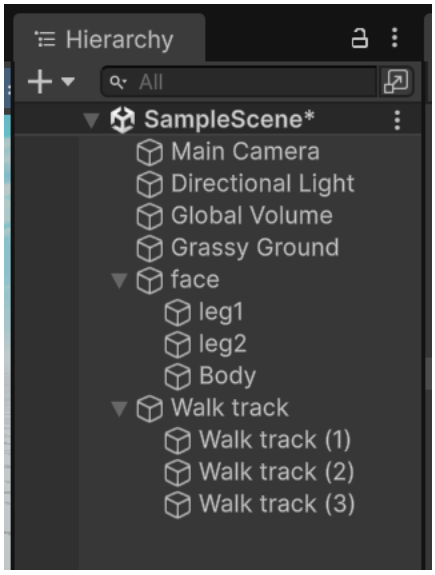
You must create a scene that includes:

• A grassy plain (ground) and a walking track

• A simple human/face structure made ONLY from Unity built in 3D objects

• Proper parent child relationships so the structure moves as one object

Hierarchy looks like:



• Application of movement and camera follow scripts (provided by instructor)

<u>Instructions:</u>

1. Project Setup

- Create a new 3D Unity project

- Create and save a new Scene

- Add a Skybox package

- Capture one screenshot showing scene creation

2. Scene Design

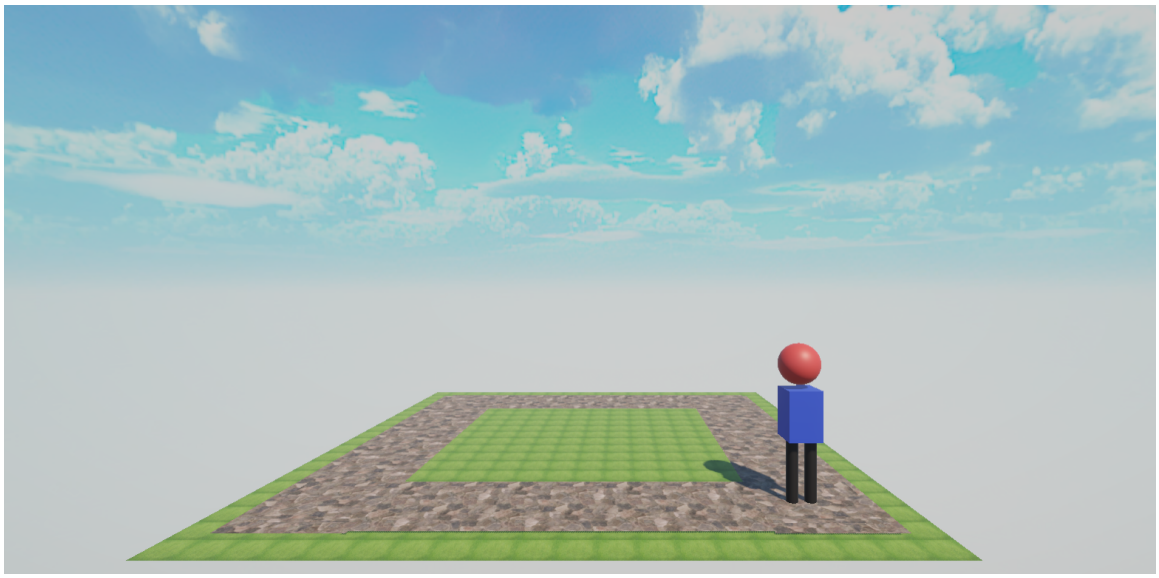
- Use only Unity primitive 3D objects (Cube, Sphere, Capsule, Cylinder, Plane, etc.)

- Design grassy ground and walking track

- Construct a human/face structure using multiple primitives

- Apply Parent Child relationships in Hierarchy to move as one object

(Sample screeshot attach)



3. Materials and Textures

- Create materials and rename them properly

- Apply colors or textures

- Adjust tiling where applicable

4. Physics Interaction

- Use Rigidbody or Collider if needed

- Adjust gravity interaction

- Modify bounciness using Physics Materials where appropriate

5. Scripts

- Attach provided Movement Script to the constructed object

- Attach Camera Follow Script to camera

- Demonstrate right, left, forward, backward movement on track, clearly show in video

6. Demonstration

- Run scene in Play Mode

- Record video showing movement and camera follow

- Include screenshots of:

  - Scene view

  - Hierarchy view

  - Script attachment

Submission Instructions

<u>Important note:</u>

Paste screenshots of your game and CS scripts in provided doc file and upload as yourReg#_filename.

Also upload 10 seconds video of the game in submission.

Copy all screenshots, CS Scripts and video in one Doc file and submit that file at google classroom Assignment_1 folder, as there is option to upload multiple files.

DO NOT UPLOAD PROJECT ZIP FILE.